Analysis II Name: ________________

Computer Science BSc Code: ________________

November 4, 2022 Group: ________________

Test 1

1. Discuss the continuity of the following function. *(9 points)*

$$f(x) = \begin{cases} \dfrac{\sin(2x-4)}{4x-8}, & x \le 0 \text{ or } x > 2 \\ \dfrac{x^2-x-2}{x^2-2x}, & 0 < x < 2 \\ 1, & x = 2 \end{cases}$$

2. Consider the function: *(6 points)*

$$f(x) = \sin(x^2) - x^3 \cdot e^{4x} + 1 \qquad (x \in \mathbb{R})$$

Prove that the equation $f(x) = 0$ has at least one real solution. Determine the equation of the tangent line to the graph of f at the point $a = 0$.

3. Discuss the following function and sketch its graph. *(11 points)*

$$f(x) = \frac{2x}{3+x^2} \qquad (x \in \mathbb{R})$$

4. Determine the following limit. *(6 points)*

$$\lim_{x \to 2} \frac{\sin(x^3 - 3x^2 + 4)}{x^2 - 4x + 4}$$

5. Find the second degree Taylor polynomial of f centered at $a = 1$ and estimate the error of approximation on the interval I. *(8 points)*

$$f(x) = \ln(x+2) \quad (x > -2), \qquad I = \left(\frac{1}{2}, \frac{3}{2}\right)$$